

## CERES, Argentina

WMO: 872570

Lat: 29.8756S

Lon: 61.9361W

Elev: 88

StdP: 100.27

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 1.1        | 2.8 | -4.9                          | 2.5 | 8.1  | -2.8 | 3.0 | 9.6  | 15.3                  | 19.4 | 12.5 | 13.8 | 1.0                   | 0    | 0.418 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 11.7                   | 36.6            | 24.0 | 35.0 | 23.8 | 33.6 | 23.3 | 26.7                | 32.5 | 26.0 | 31.6 | 25.2 | 30.6 | 5.7                  | 0    |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 25.3                            | 20.7 | 29.7 | 24.5 | 19.7 | 28.9 | 23.7 | 18.7 | 28.1 | 84.8          | 32.3 | 81.2 | 31.7 | 78.0 | 30.6 | 30.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 12.4          | 11.6 | 9.1 | -1.9                       | 39.2 | 1.5                | 1.8 | -3.0                                               | 40.5 | -3.9       | 41.6 | -4.8       | 42.6 | -5.9       | 43.9 |    |    |
| (5)               |      |     | -2.7                       | 28.3 | 1.5                | 1.0 | -3.8                                               | 29.0 | -4.7       | 29.6 | -5.5       | 30.2 | -6.6       | 31.0 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 19.6   | 25.8  | 24.8  | 22.8  | 19.6  | 16.3  | 13.5  | 12.9  | 14.8  | 17.2  | 20.3  | 22.4  | 24.6  |  |  |
|                                                                        | DBStd       | 5.81   | 2.92  | 3.10  | 3.14  | 3.83  | 4.01  | 4.20  | 4.54  | 4.80  | 4.40  | 3.77  | 3.21  | 3.05  |  |  |
|                                                                        | HDD10.0     | 39     | 0     | 0     | 0     | 0     | 1     | 12    | 17    | 8     | 2     | 0     | 0     | 0     |  |  |
|                                                                        | HDD18.3     | 693    | 0     | 1     | 3     | 29    | 91    | 156   | 180   | 135   | 73    | 21    | 4     | 1     |  |  |
|                                                                        | CDD10.0     | 3529   | 490   | 414   | 397   | 288   | 196   | 117   | 108   | 156   | 216   | 319   | 373   | 453   |  |  |
|                                                                        | CDD18.3     | 1141   | 232   | 182   | 143   | 67    | 27    | 11    | 13    | 24    | 37    | 82    | 128   | 196   |  |  |
|                                                                        | CDH23.3     | 11368  | 2529  | 1789  | 1234  | 502   | 164   | 51    | 89    | 281   | 443   | 857   | 1343  | 2087  |  |  |
|                                                                        | CDH26.7     | 4953   | 1247  | 813   | 483   | 163   | 37    | 9     | 22    | 110   | 167   | 347   | 579   | 977   |  |  |
| (14) Wind                                                              | WSAvg       | 3.6    | 3.3   | 3.3   | 3.1   | 3.2   | 3.0   | 3.2   | 3.6   | 4.1   | 4.4   | 4.4   | 4.0   | 3.6   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 928    | 137   | 114   | 139   | 89    | 29    | 20    | 15    | 17    | 40    | 80    | 106   | 136   |  |  |
|                                                                        | PrecMax     | 1408   | 420   | 387   | 396   | 302   | 110   | 127   | 105   | 89    | 152   | 230   | 275   | 450   |  |  |
|                                                                        | PrecMin     | 450    | 13    | 5     | 6     | 2     | 0     | 0     | 0     | 0     | 0     | 4     | 16    | 15    |  |  |
|                                                                        | PrecStd     | 215    | 92    | 69    | 85    | 66    | 28    | 24    | 21    | 19    | 35    | 52    | 55    | 93    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 38.5  | 37.7  | 35.4  | 33.2  | 30.3  | 27.9  | 29.8  | 33.4  | 35.3  | 36.2  | 36.3  | 38.6  |  |  |
|                                                                        |             | MCWB   | 24.7  | 24.7  | 24.8  | 23.7  | 22.6  | 20.4  | 19.7  | 21.1  | 21.3  | 22.5  | 23.0  | 24.7  |  |  |
|                                                                        | 2%          | DB     | 36.1  | 35.4  | 33.3  | 30.8  | 27.4  | 24.2  | 25.9  | 29.8  | 31.2  | 32.8  | 34.2  | 35.8  |  |  |
|                                                                        |             | MCWB   | 24.3  | 24.8  | 23.9  | 22.6  | 21.0  | 19.7  | 19.1  | 19.7  | 19.8  | 22.0  | 21.8  | 23.7  |  |  |
|                                                                        | 5%          | DB     | 34.4  | 33.4  | 31.4  | 28.6  | 25.0  | 21.9  | 22.8  | 26.3  | 28.4  | 30.4  | 32.1  | 33.7  |  |  |
|                                                                        |             | MCWB   | 23.8  | 24.1  | 23.0  | 21.5  | 20.3  | 18.0  | 16.9  | 18.1  | 18.7  | 20.6  | 21.3  | 23.4  |  |  |
|                                                                        | 10%         | DB     | 32.7  | 31.5  | 29.6  | 26.2  | 22.8  | 19.9  | 20.2  | 23.1  | 25.7  | 28.0  | 30.2  | 31.8  |  |  |
|                                                                        |             | MCWB   | 23.4  | 23.4  | 22.3  | 20.8  | 18.8  | 16.1  | 15.3  | 16.4  | 17.2  | 19.4  | 20.8  | 22.7  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 27.9  | 27.8  | 26.8  | 25.6  | 24.4  | 21.9  | 21.4  | 22.4  | 23.2  | 25.6  | 25.8  | 27.4  |  |  |
|                                                                        |             | MCDB   | 34.0  | 33.6  | 32.8  | 30.3  | 28.1  | 25.0  | 27.1  | 30.3  | 30.7  | 32.2  | 32.8  | 33.1  |  |  |
|                                                                        | 2%          | WB     | 26.6  | 26.6  | 25.7  | 24.2  | 22.5  | 20.4  | 19.5  | 20.6  | 21.5  | 23.6  | 24.2  | 26.1  |  |  |
|                                                                        |             | MCDB   | 32.3  | 32.0  | 30.5  | 28.3  | 25.8  | 23.4  | 24.5  | 27.9  | 28.6  | 29.9  | 30.2  | 31.8  |  |  |
|                                                                        | 5%          | WB     | 25.8  | 25.6  | 24.6  | 23.1  | 20.9  | 18.9  | 18.0  | 19.0  | 20.0  | 22.3  | 23.0  | 25.1  |  |  |
|                                                                        |             | MCDB   | 31.5  | 31.0  | 28.7  | 26.6  | 24.0  | 21.4  | 21.9  | 25.3  | 26.3  | 28.1  | 28.8  | 31.0  |  |  |
|                                                                        | 10%         | WB     | 24.9  | 24.7  | 23.6  | 21.8  | 19.5  | 17.2  | 16.5  | 17.2  | 18.6  | 20.9  | 21.9  | 24.0  |  |  |
|                                                                        |             | MCDB   | 30.1  | 29.7  | 27.6  | 24.9  | 22.2  | 19.2  | 19.6  | 22.2  | 23.9  | 26.1  | 27.4  | 29.1  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.7  | 10.9  | 10.8  | 10.0  | 9.8   | 10.1  | 11.6  | 13.1  | 13.0  | 12.2  | 12.6  | 12.0  |  |  |
|                                                                        |             | MCDBR  | 14.1  | 13.7  | 13.3  | 12.9  | 11.8  | 11.6  | 14.0  | 17.0  | 17.0  | 15.9  | 16.0  | 15.0  |  |  |
|                                                                        | 5% WB       | MCWBR  | 4.7   | 5.2   | 5.3   | 5.9   | 6.1   | 6.5   | 7.3   | 8.1   | 7.6   | 7.0   | 6.2   | 5.7   |  |  |
|                                                                        |             | MCWBR  | 11.2  | 11.1  | 10.5  | 10.1  | 9.8   | 9.4   | 11.6  | 14.9  | 13.5  | 13.1  | 12.5  | 12.0  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.406 | 0.397 | 0.385 | 0.382 | 0.361 | 0.360 | 0.362 | 0.433 | 0.497 | 0.449 | 0.403 | 0.400 |  |  |
|                                                                        | taud        |        | 2.375 | 2.407 | 2.434 | 2.415 | 2.431 | 2.415 | 2.379 | 2.125 | 1.958 | 2.177 | 2.334 | 2.375 |  |  |
|                                                                        | Ebn at Noon |        | 936   | 926   | 904   | 852   | 821   | 795   | 809   | 782   | 782   | 867   | 933   | 945   |  |  |
|                                                                        | Edh at Noon |        | 130   | 122   | 112   | 104   | 92    | 88    | 95    | 136   | 178   | 152   | 135   | 131   |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 6.96  | 6.20  | 5.16  | 3.88  | 2.99  | 2.63  | 3.05  | 4.01  | 4.99  | 5.83  | 6.90  | 7.07  |  |  |
|                                                                        | RadStd      |        | 0.48  | 0.40  | 0.38  | 0.48  | 0.28  | 0.31  | 0.28  | 0.31  | 0.47  | 0.45  | 0.41  | 0.39  |  |  |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           |  | +0.53 | +0.78   | -1.62  | +0.74   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (0 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page